DT → random forest

① n_est = [100, 200, 300, 400]

② max depth = [3, 5, 7, 10]

Grid Search CV

↳ automated hyper-parameter tuning.

3 { n_1, n_2 } × 4 { m_1, m_2, m_3, m_4 } = 4×4

→ too many models (exhaustive) ↳ brute force (check for all values)

4×4 = 16 models

hyper parameters n_1 3, n_2 4, n_3 2, n_4 1 → 3×4×2×1 = 24 models

CV = 3

24×3 = models

each 6 every time 3 models

CV=3

Random Search CV

100, 7 → 150, 125 → 350

6, 8

→ CCP - alpha

base learners → high depth ↳ high variance

→ optimal complexity

Pruning

obj function = min [α (with high wrong prediction) + Δ (with high complexity)]

obj function ⇒ min [α (outlier impact) + α # leaf nodes]

α ↑ → underfit

α ↓ → overfit

Ensemble → Take some base learners (multiple models) → combine the output

RF: DT + RS + CS → agg

extremely randomize Trees

↳ extra tree classifier

(M.L) Classic learners → dt = 5 ⇒ 1000 samples → 200 → Threshold → 0.42 → 0.42 best

BOOSTING

Bagging: Base models { high var + low bias } + Randomness + agg → reduce var

Boosting: Base learners { high bias + low var } + additive combining → reduce bias

'under fit'

Core Idea: → reduce the BIAS

Task → Regression

$D_{train} \Rightarrow \{ x^{(i)}, y^{(i)} \}_{i=1}^{n}$ ∈ R

[Step = 0]

(a) D_{train} → m_0 → simple model (high bias, low variance) $d_{max} = 1$

$F_0 \Rightarrow y = h_0(x)$ → average model

(b) error / residual

$error^{(i)} = y^{(i)} - h_0(x^{(i)})$ (simple dt)

$\{ x^{(i)}, y^{(i)}, error^{(i)} \}_{i=1}^{n}$

$x^{(i)} \Rightarrow y^{(i)}$ (actual) $y^{(i)} + error^{(i)}$

step(1) $\{ x^{(i)}, error^{(i)} \}_{i=1}^{n}$ → $h_1(x)$ (weak)

some kind of weight → $F_1(x) = h_0(x) + \alpha_1 h_1(x)$ → error of previous model

step2 $\{ x^{(i)}, err^{(i)}_{2} \}_{i=1}^{n}$ → h_2 (weak) → $h_2(x)$

$F_2(x) = h_0(x) + \alpha_1 h_1(x) + \alpha_2 h_2(x)$

⋮

step m $F_m(x) = h_0(x) + \alpha_1 h_1(x) + \alpha_2 h_2(x) \dots \alpha_m h_m(x)$

$F_m(x) = h_0(x) + \sum_{i=1}^{m} \alpha_i h_i(x)$

$h_0(x) = 8$ (ȳ)

$y^{(i)} - h_0(x)$

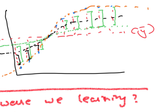
$= 12 - 8 = 4$

$f_m(x) = h_0(x) + \alpha_1 h_1(x)$ ↓ ↓

8, 1.1, 0.9, 0.8

$\hat{y} = w_0 + \sum_{j=1}^{d} w_j x_j$

Geometric Intuition



① → What were we learning?

3 trees ⇒ m

$F_3(x) = h_0(x) + \alpha_1 h_1(x) + \alpha_2 h_2(x) + \alpha_3 h_3(x)$

mean → ȳ

no. of models / trees gen created

m ↑ → overfit

m ↓ → underfit (bias ↑)

Classification

D_{train} $\{ x^{(i)}, y^{(i)} \}_{i=1}^{n}$; y ∈ {0, 1}

step(0) M_0 ⇒ (0, 1) → 0.4 → 1 → 0.4

step (1) $\{ x^{(i)}, error^{(i)} \}$

$F_1(x) = m_0 + \alpha_1 m_1$

$F_m(x) = m_0 + \alpha_1 m_1 + \alpha_2 m_2 \dots \alpha_m m_m$

$F_m(x_1) = m_0 + \alpha_1 m_1 + \alpha_2 m_2 \dots \alpha_m m_m$